Deep Learning and Image Analysis

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Exercise Content:

※ Basic training concept of deep learning for image processing

※ Convolutional Neural Networks(CNN)

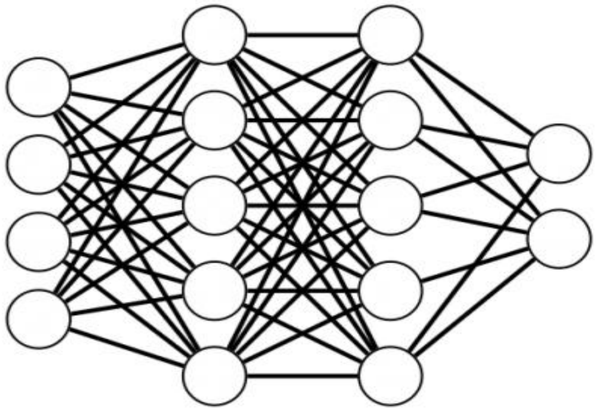
※ Google Colab tutorial for image processing(road crack classification)

What is Machine Learning?

- Computer algorithms that can improve automatically through experience and the use of data [Tom Mitchell, 1997]

- The way to achieve Artificial Intelligence (AI)

Machine learning	Main Feature
Supervised Learning	Labeled data
Unsupervised Learning	Unlabeled data
Reinforcement Learning	Guided by rewards / punishment

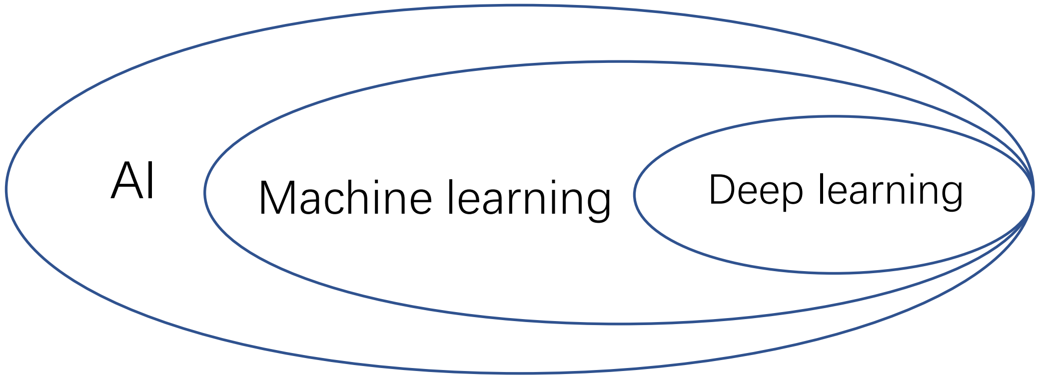


Deep learning

- A branch of Machine Learning

- Use Deep Neural Networks(DNN) that resemble the structures of human brains

- Widely used for image processing



Training steps for machine learning processing :

1.Import the Data

2.Clean the Data

3.Split the Data into Training/ Test Sets

4.Create a Model

5.Train the Model

6.Make predictions

7.Evaluate and Improve

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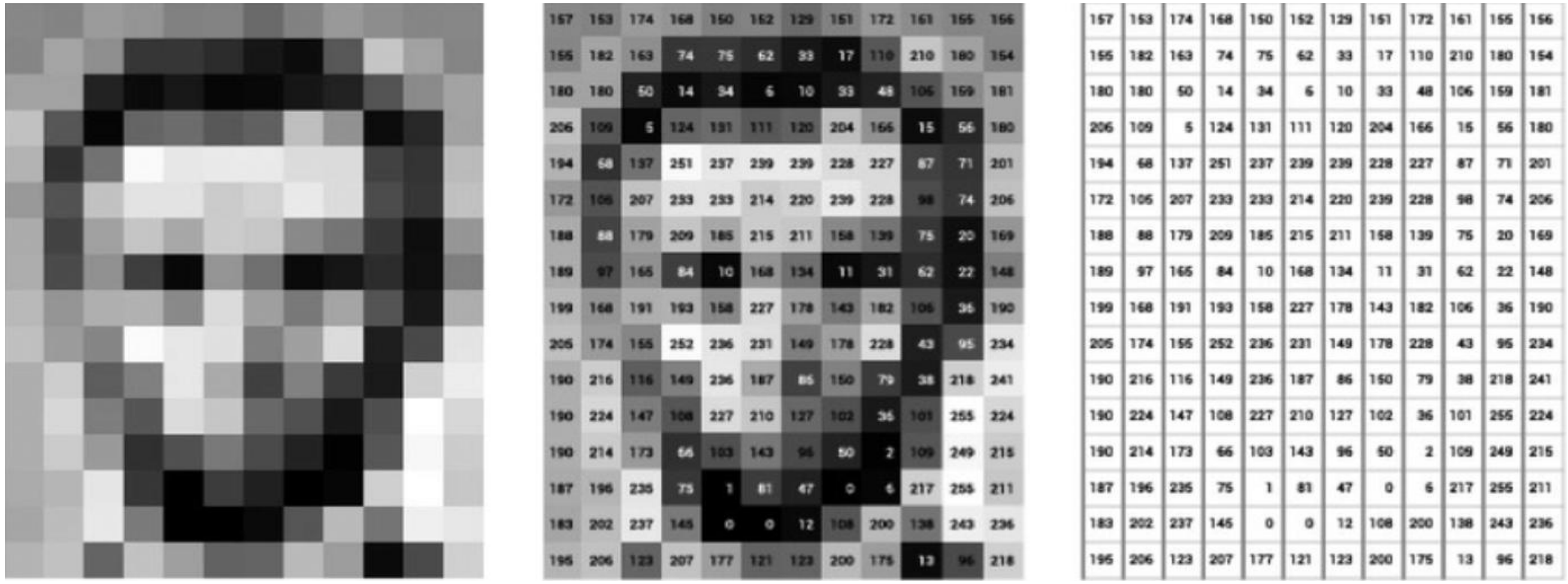
Libraries:

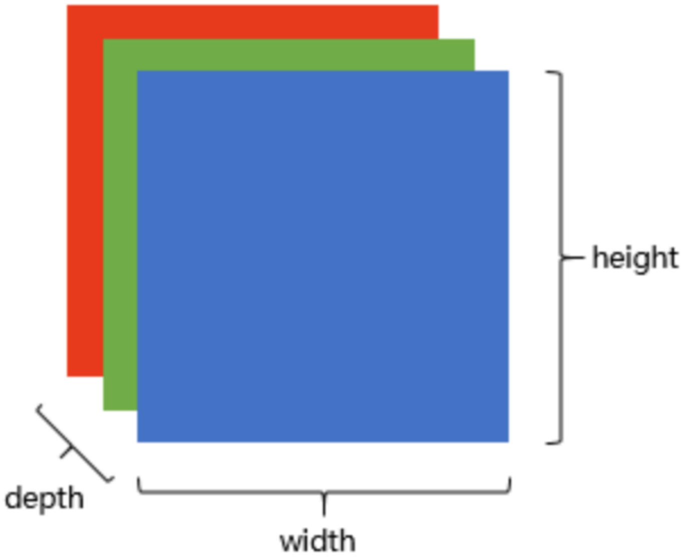
1.Numpy

2. Pandas

3.MatplotLib/cv2

Basic concept of the image:





The images are all numbers(matrix).
For grayscale image is 1 channel, color image (RGB) are 3 channels.

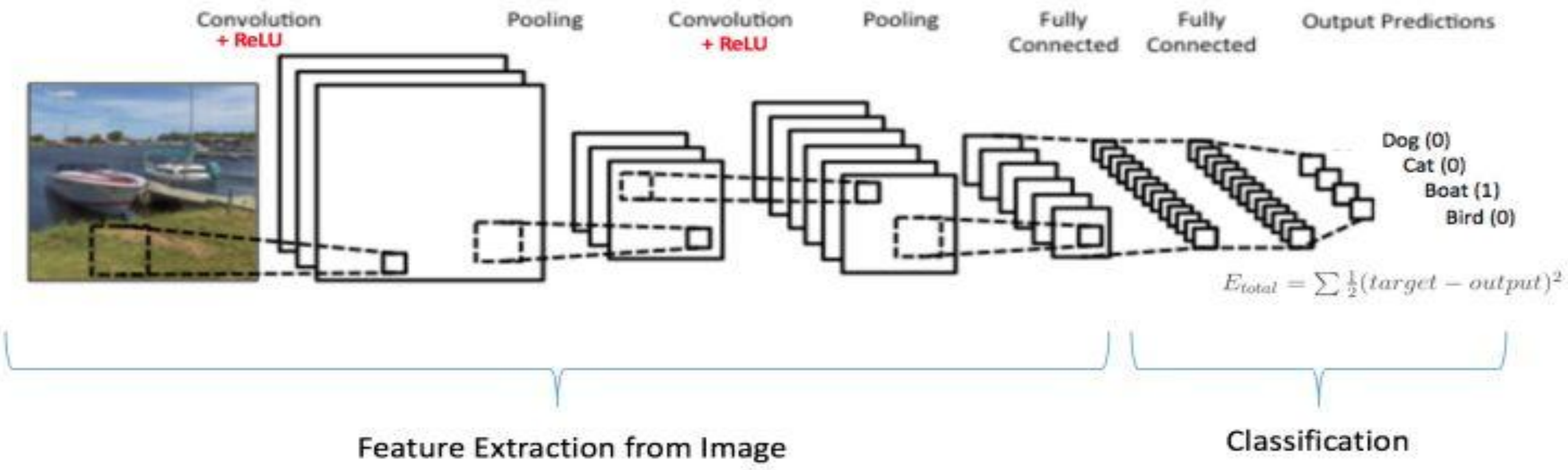
Format: The image can be saved as png/jpeg/gif.

Resolution: refers to the number of pixel in the image or it can be regarded as size of the image.

Pixel value: for 8-bit image, the pixel range is 0-255, 16-bit is 0-(2^{16}-1)

Convolutional Neural Networks(CNN):

※ CNN has several applications including image recognition.



Convolutional layer, pooling layer, ReLu layer(activation function), fully-connected layer

Input

4	9	2	5	8	3
5	6	2	4	0	3
2	4	5	4	5	2
5	6	5	4	7	8
5	7	7	9	2	1
5	8	5	3	8	4

$n_H \, x \, n_W = \, 6 \, x \, 6$

$*$

Filter

1	0	-1
1	0	-1
1	0	-1

=

Result

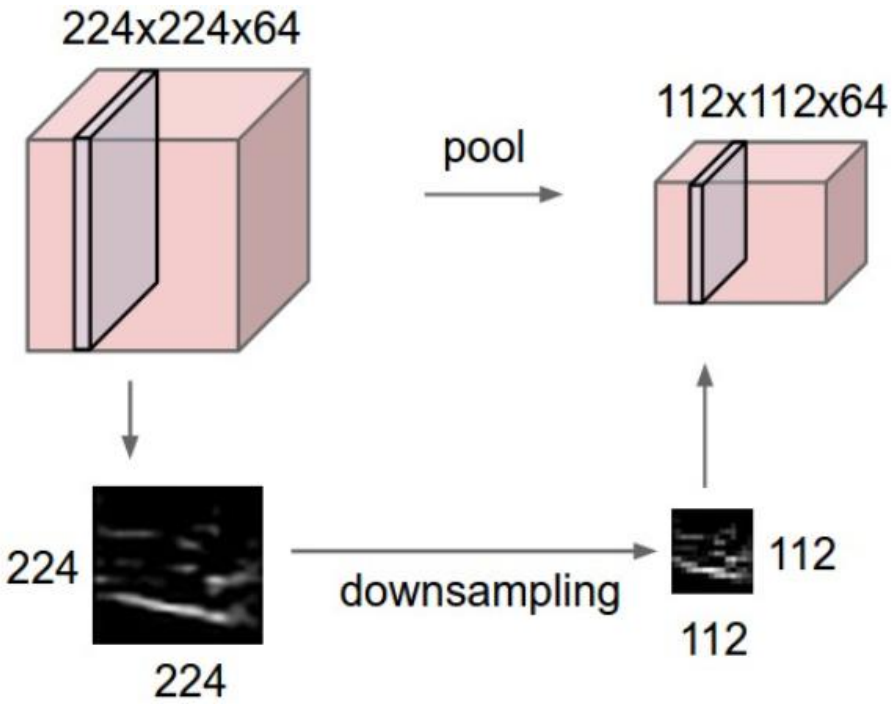
2			

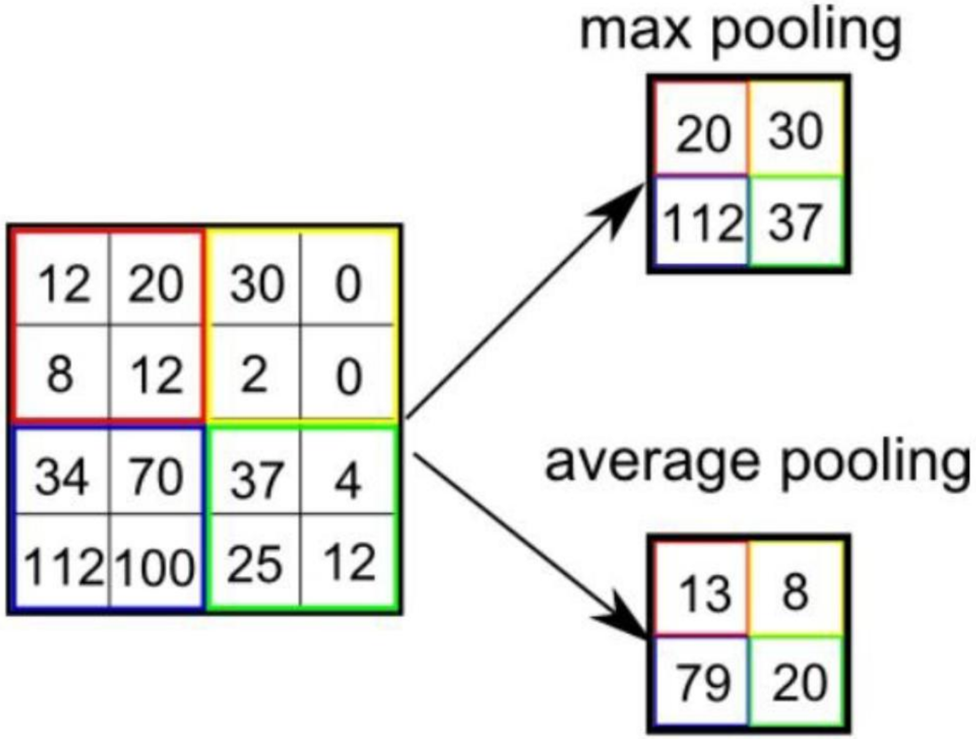
Parameters:

Size: $f = 3$
Stride: $s = 1$
Padding: $p = 0$

$\boxed{2}$ = 4*1 + 9*0 + 2*(-1) +
5*1 + 6*0 + 2*(-1) +
2*1 + 4*0 + 5*(-1)

Pooling:

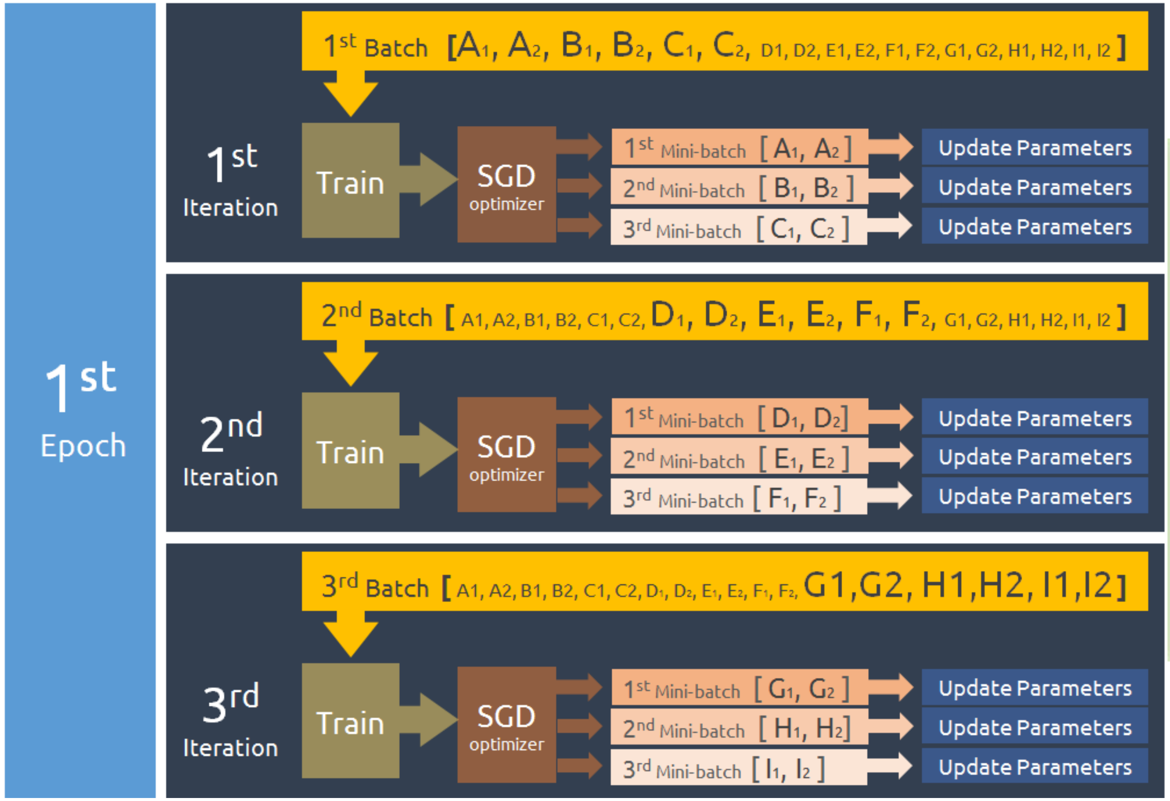




Max pooling   Average pooling

Reduce the volume spatially, speed up the training

Model training:



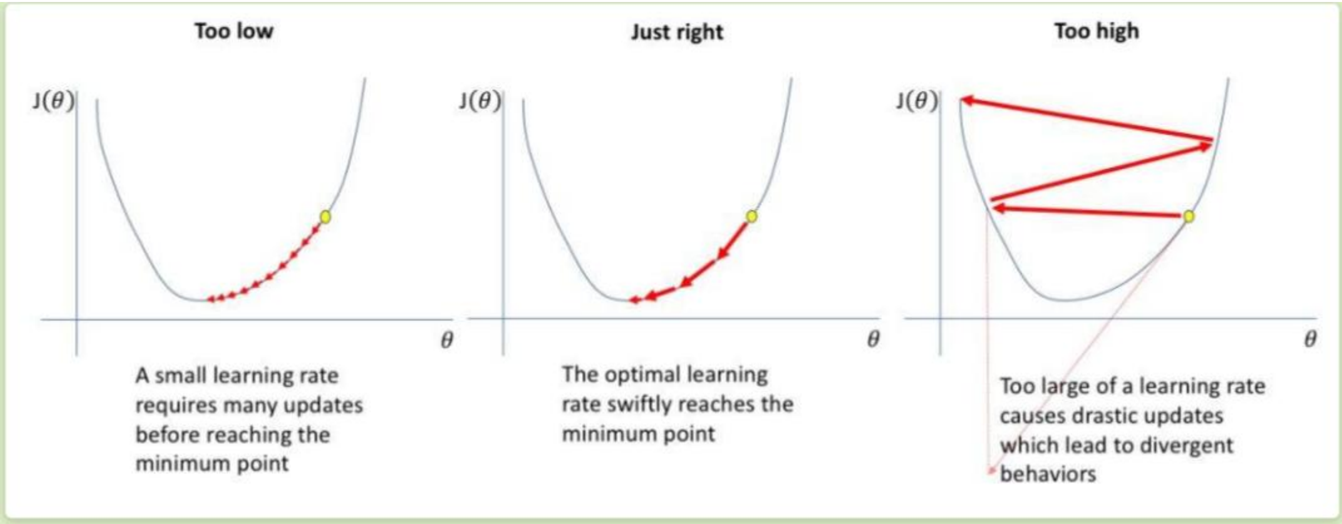
Sample: 5,000 images

Batch_size= 10

Iteration= 500

Epoch=1=500 iterations

Learning_rate: the speed of learning



The essence of training model is to get the function!

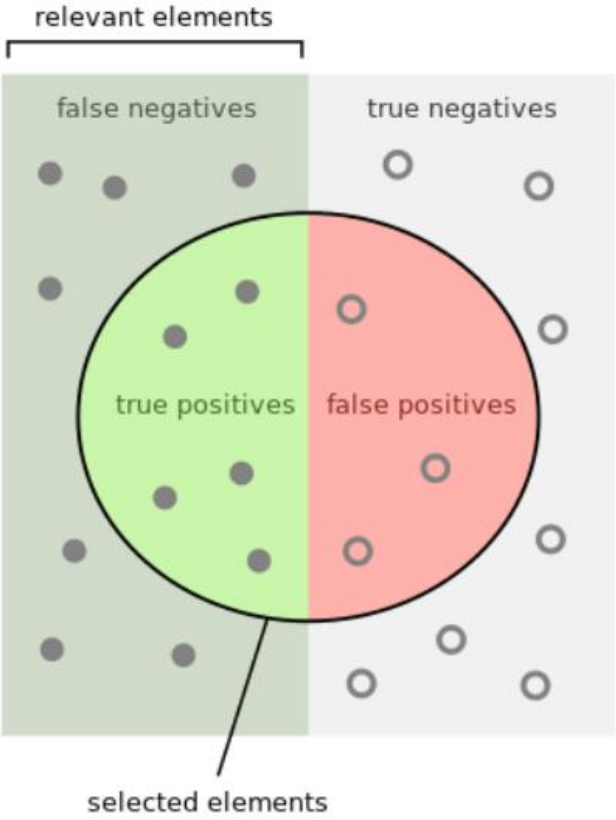
$$x \ggg \boxed{f(x)} \ggg y$$

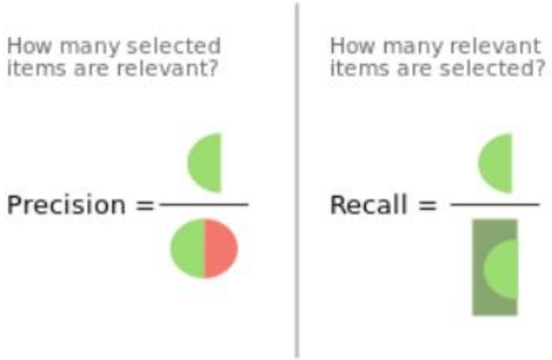
input model output

f(x)=w1∗x1+w2∗x2+⋯+wd∗xd+b

We want to get the most optimized model to generates accurate predictions for not-yet-seen data!

How to evaluate your model:





		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

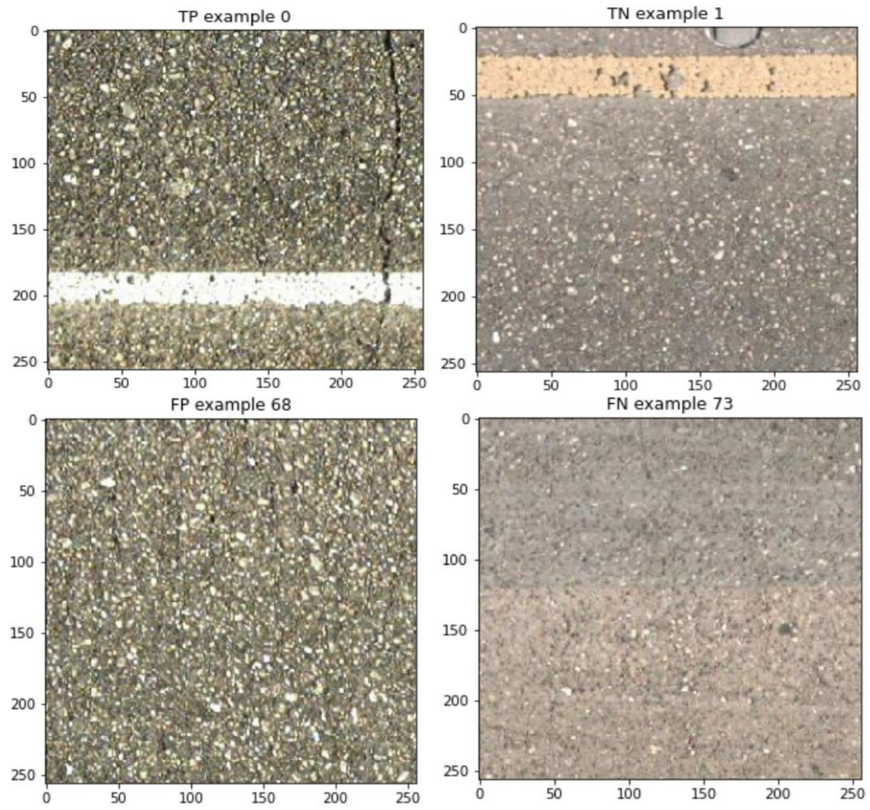
$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall}$$

$$accuracy = \frac{TP + TN}{TP + FN + TN + FP}$$

$$specificity = \frac{TN}{TN + FP}$$



TP: the image with crack was predicted as positive

TN: the image without damage was predicted correctly

FP: the image without crack was regarded as cracked image

FN: the image with crack was predicted as no cracked

Recommended Reading: Deep Learning & Image Processing in Advanced Research

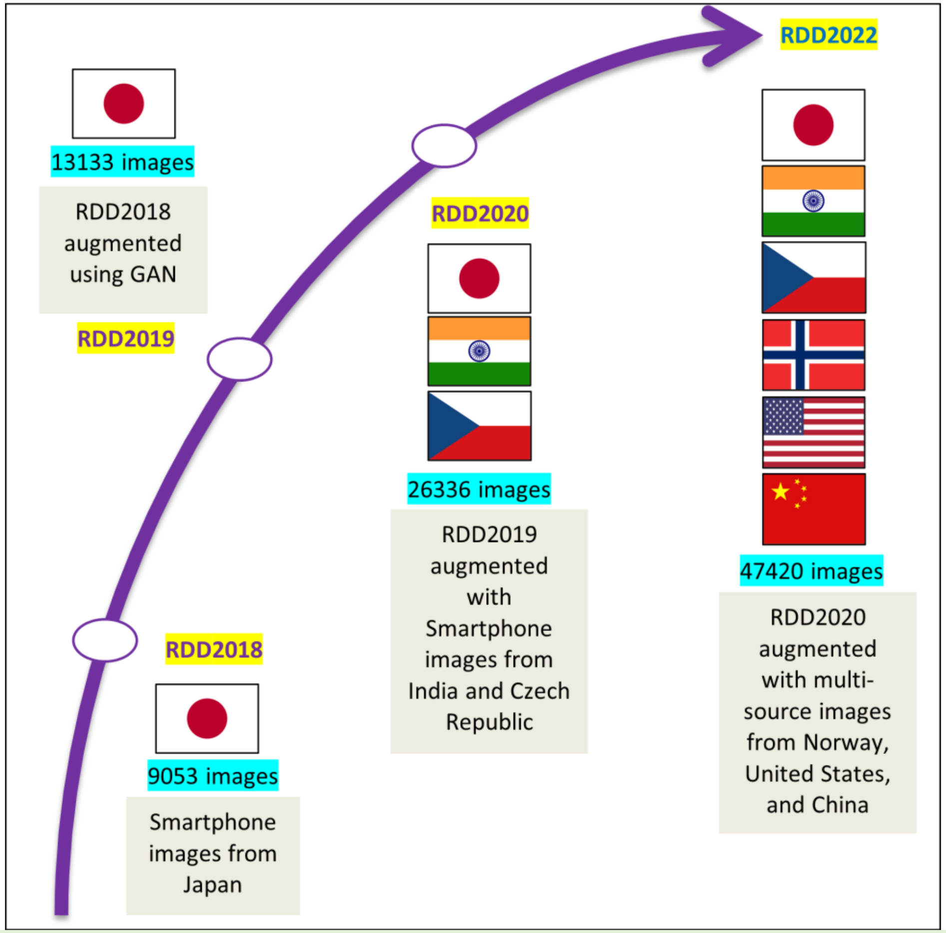
AI-driven Global Road Damage Detection

Started at UTokyo, this multinational project has scaled globally through Open-Access Data and Big Data Cup challenges, driving AI-powered infrastructure monitoring.

Useful Links:

1. Data Download links: 2018, 2019, 2020, 2022
2. Data Cup Summary papers: 2020, 2022, 2024
3. Related Analysis Papers: 2021, 2024, 2025
4. Data Articles: 2021, 2022, 2024
5. Related Research Articles: RDD2018, RDD2019, RDD2020, RDD2022
6. GitHub Website for latest news and updates

Global spread of participants in the latest data cup ORDDC'2024



Open Access Datasets released through the project. The latest version, RDD2022 has been downloaded over 22,000 times (since 2022).